National University of Computer and Emerging Sciences



Lab Manual 07
Computer Organization
and Assembly Language

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Objectives:

- Understand and apply MUL and DIV instructions
- Subroutines with Stack, Local Variables and strings
- Display Memory and Screen Location Calculation

Activities:

Activity 1 (Assembly Language Tasks):

1) *Sum of two numbers via stack*

   Write a program that defines a basic stack function to calculate the sum of two numbers. The function should accept two integers as parameters, passed through the stack. The function should return the sum in AX. Use stack pointer registers, subroutines and push/pop to perform addition.

2) *Calculating Power*

   Write a subroutine Power that takes x and y as parameters, calculates x raise to the power y, and returns the result. Use MUL instruction for multiplication and test your function with three different data/function calls.

3) *Square using stack*

   Write a subroutine that computes the square of a number passed as a parameter and returns the result through the stack. The main program should call the subroutine and pop the result into a memory variable.

4) *Factorial using local variables*

   Write a subroutine to compute the factorial of a number using the stack. The main program should pop the result into memory. The subroutine should:
   - Accept n as input (pushed by the main program).
   - Use the stack to store intermediate values (local variables).
   - Push the result on the stack.

5) *Transpose of a matrix*

   A transpose of a matrix is actually a matrix that is formed by turning all the rows of a given matrix into columns and vice versa.

   Suppose you have an array representing a 2D Matrix placed in a row-major order along with its rows and columns as

   arr: dw 1 , 2
   dw 3 , 4
   dw 5 , 6
   dw 7 , 8
   rows: dw 4
   columns: dw 2

   Write a subroutine that will be passed 3 parameters
   - Address of this array
   - Address of rows
   - Address of columns

The subroutine must implement a transpose of that matrix (represented by an array) without using/allocating any extra array and update rows and columns respectively. (You may use 1 or 2 local variables, though, only if you need them).

After the subroutine execution is complete, the arr will become:

arr 1 , 3 , 5, 7
2 , 4 , 6, 8
rows 2
columns 4

Remember, this is just a representation it can also be represented like this
arr: dw 1, 2, 3, 4, 5, 6, 7, 8
arr: dw 1, 3, 5, 7, 2, 4, 6, 8 (final result)

6) *Define Characters*

Define 5 characters in memory ('A', 'B', 'C', 'D', 'E').
Write a program to move them into registers one by one and store them in another memory location.

7) *Copy String*

Define a string msg db 'Assembly Rocks$',0.
Write a program to copy this string into another memory buffer using a loop.

8) *String Reverse*

Write a program to reverse a string defined in memory.
Example: "HELLO$" → "OLLEH$".
Store reversed string in a new memory buffer.

9) *Printing Rectangle*

Write a function PrintRectanlge that prints a rectangle having its TopLeft and BottomRight corners at (top,left) and (bottom,right) coordinates respectively where top, left, bottom and right are parameters passed by caller. Also pass attribute by caller to print colored rectangle. Following is a red rectangle with TopLeft = (2, 10) and BottomRight = (20, 60).